



NATIONAL AEROSPACE STANDARD

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FED. SUPPLY CLASS

5305

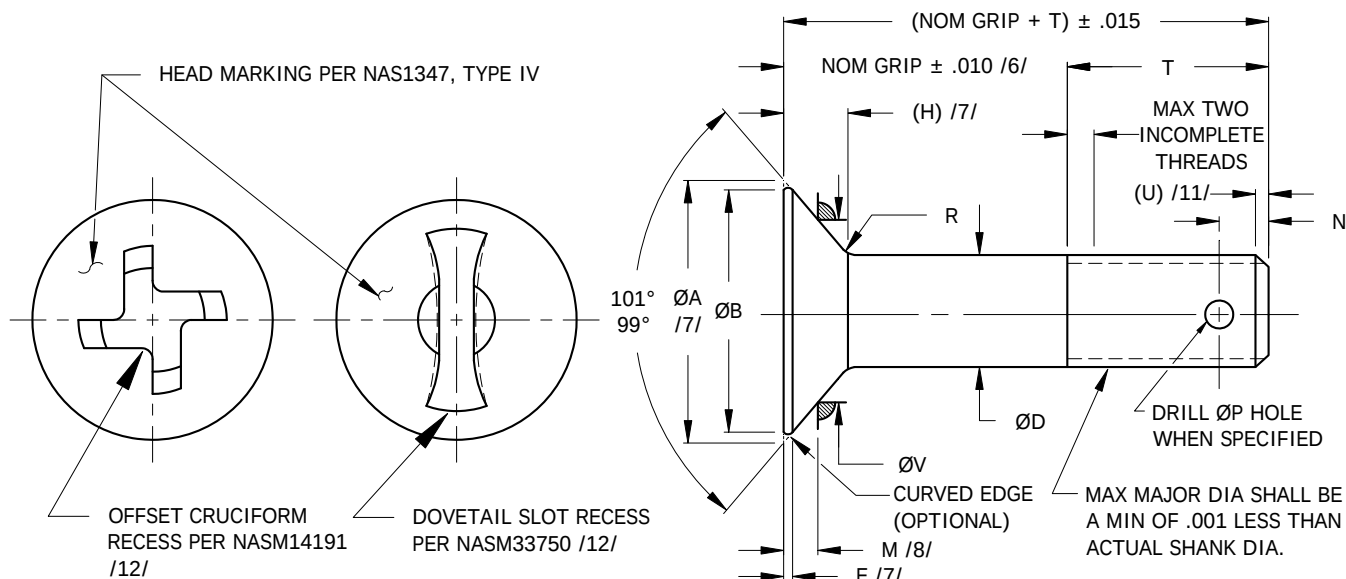


FIGURE 1 – 100° FLUSH HEAD BOLT

TABLE I – DIMENSIONS

FIRST DASH NO.	THREAD UNJF-3A	ØA MAX /7/	ØB MIN	ØD		E MAX /7/	(H) /7/	M GAGE PROT /8/	N	ØP	R RAD	T	(U) /11/	ØV GAGE	Y FIM /9/	Z FIM /10/
				PLATED	UN-PLATED											
3	.1900-32	.381	.339	.1895 .1885	.1895 .1890	.015	.083	.0285 .0255	.126 .106	.080 .070	.035 .015	.338	.050	.3147 .3145	.005	.0040
4	.2500-28	.506	.464	.2495 .2485	.2495 .2490	.015	.111	.0338 .0309	.126 .106	.086 .076	.035 .015	.425	.050	.4245 .4243	.006	.0030
5	.3125-24	.634	.578	.3120 .3110	.3120 .3115	.015	.140	.0409 .0370	.130 .110	.086 .076	.045 .025	.469	.060	.5389 .5387	.007	.0030
6	.3750-24	.760	.717	.3745 .3735	.3745 .3740	.015	.167	.0459 .0422	.130 .110	.116 .106	.045 .025	.578	.060	.6532 .6530	.008	.0025
7	.4375-20	.889	.826	.4370 .4360	.4370 .4365	.022	.195	.0470 .0425	.134 .114	.116 .106	.055 .035	.694	.080	.7784 .7782	.009	.0025
8	.5000-20	1.014	.951	.4995 .4985	.4995 .4990	.022	.222	.0529 .0482	.134 .114	.116 .106	.055 .035	.735	.080	.8902 .8900	.010	.0020
9	.5625-18	1.141	1.071	.5615 .5605	.5615 .5610	.025	.250	.0590 .0541	.134 .114	.151 .141	.055 .035	.840	.090	1.0028 1.0026	.010	.0020
10	.6250-18	1.272	1.201	.6240 .6230	.6240 .6235	.025	.280	.0683 .0631	.134 .114	.151 .141	.055 .035	.902	.090	1.1124 1.1122	.010	.0020

THIRD ANGLE PROJECTION	
PROCUREMENT SPECIFICATION	NOTED

CUSTODIAN	NATIONAL AEROSPACE STANDARDS COMMITTEE
TITLE	BOLT, 100° FLUSH HEAD, 160 KSI UTS

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CLASSIFICATION PART STANDARD	NAS1790
	SHEET 1 OF 6

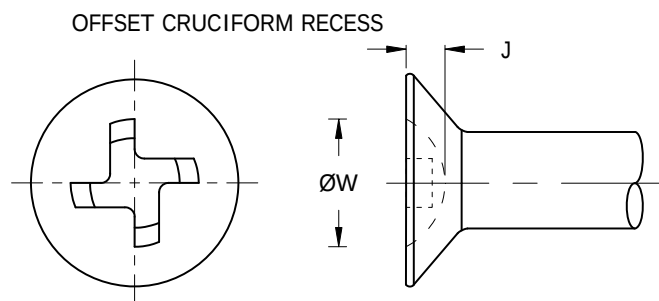


FIGURE 2 – OFFSET CRUCIFORM

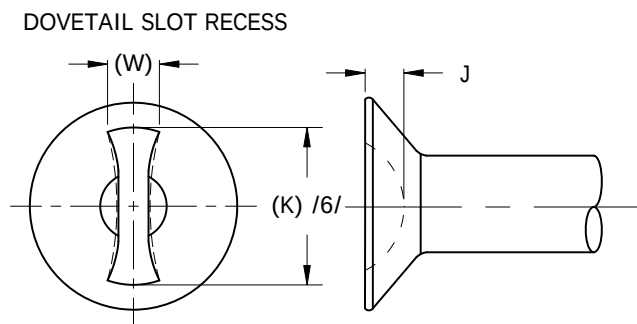


FIGURE 3 – DOVETAIL SLOT RECESS

TABLE II – OFFSET CRUCIFORM DIMENSIONS

FIRST DASH NO.	RECESS NO.	J MAX	ØW MAX	GAGE PENETRATION		RECESS VALUES	
				MIN	MAX	TORQUE IN-LBF MIN	RAISED METAL IN. MAX
3	10	.090	.248	.0560	.0685	60	.005
4	1/4	.118	.325	.0750	.0890	140	.005
5	5/16	.122	.357	.0700	.0860	220	.005
6	3/8	.145	.427	.0850	.1030	520	.006
7	7/16	.169	.498	.1005	.1205	750	.007
8	1/2	.193	.568	.1155	.1375	900	.008
9	9/16	.217	.638	.1305	.1545	1,300	.009
10	5/8	.241	.708	.1450	.1710	2,100	.010

TABLE III – DOVETAIL SLOT RECESS DIMENSIONS

FIRST DASH NO.	RECESS NO.	J		(K)	(W)	RECESS VALUES	
		MIN	MAX			TORQUE IN-LBF MIN	RAISED METAL IN. MAX
3	3	.042	.045	.305	.110	60	.005
4	4	.051	.054	.422	.134	140	.005
5	5	.061	.064	.490	.160	220	.005
6	6	.084	.088	.639	.198	520	.006
7	7	.096	.100	.715	.224	750	.007
8	8	.101	.105	.766	.256	900	.008
9	9	.114	.118	.877	.300	1,300	.009
10	10	.137	.141	1.020	.344	2,100	.010

MATERIAL:

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ALLOY STEEL: 4140 (UNS G41400) PER AMS6382, 4340 (UNS G43406) PER AMS6415 OR AMS6484, OR 8740 (UNS G87400) PER AMS6322, AMS6325, OR AMS6327.

CRES: A286 (UNS S66286) CONFORMING TO THE CHEMISTRY OF AMS5731, AMS5732, AMS5737 OR AMS5853.

TITANIUM ALLOY: 6Al-4V (UNS R56400) PER AMS4928 OR AMS4967.

HEAT TREATMENT:

DEVELOP BASIC MATERIAL PROPERTIES AS FOLLOWS, WITH CONTROLS PER AMS2759 FOR ALLOY STEEL AND CRES AND AMS-H-81200 OR AMS2801 FOR TITANIUM:

ALLOY STEEL: 160-180 KSI Ft_u.

CRES: 160-190 KSI Ft_u. PRECIPITATION HARDENING PER AMS5853. /4/

TITANIUM (6Al-4V): 160 KSI Ft_u PER AMS4928 OR AMS4967.

FINISH:

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CADMIUM PLATE PER AMS-QQ-P-416, TYPE II, CLASS 2;

PASSIVATE PER AMS2700, METHOD 1, CLASS 4;

PASSIVATE PER AMS2700, METHOD 1, TYPE 8, CLASS 4 - NO POST TREATMENT;

PASSIVATE PER AMS2700, METHOD 2 - NO POST TREATMENT;

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ALUMINUM COAT PER NAS4006, NO CLASS;

ALUMINUM COAT PER NAS4006, CLASS NC;

OR NO FINISH.

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LUBRICATION:

DRY FILM LUBRICANT PER ~~MIL-PRF-46010~~ AS5272, TYPE I.

CODE:

⑥ FIRST DASH NUMBER INDICATES NOMINAL DIAMETER IN .0625 INCH INCREMENTS. SECOND DASH NUMBER INDICATES GRIP LENGTH IN .0625 INCH INCREMENTS.

MATERIAL CODE FOLLOWS BASIC PART NUMBER:
NO CODE LETTER INDICATES ALLOY STEEL.
“C” INDICATES CRES.
“V” INDICATES TITANIUM.

FINISH CODE FOLLOWS SECOND DASH NUMBER:

ALLOY STEEL:

NO CODE LETTER INDICATES TYPE II, CLASS 2 CADMIUM PLATE.
“A” INDICATES ALUMINUM COAT PER NAS4006, NO CLASS.
“F” INDICATES ALUMINUM COAT PER NAS4006, CLASS NC.

CRES:

⑥ NO CODE LETTER INDICATES PASSIVATE PER AMS2700, METHOD 1, CLASS 4.
“A” INDICATES ALUMINUM COAT PER NAS4006, NO CLASS.
“B” INDICATES PASSIVATE PER AMS2700, METHOD 2 - NO POST TREATMENT;
“F” INDICATES ALUMINUM COAT PER NAS4006, CLASS NC.
⑥ “M” INDICATES PASSIVATE PER AMS2700, METHOD 1, TYPE 8, CLASS 4 - NO POST TREATMENT
“P” INDICATES TYPE II, CLASS 2 CADMIUM PLATE.

TITANIUM:

⑥ NO CODE LETTER INDICATES NO FINISH.
“A” INDICATES ALUMINUM COAT PER NAS4006, NO CLASS.
“F” INDICATES ALUMINUM COAT PER NAS4006, CLASS NC.

RECESS CODE FOLLOWS FIRST DASH NUMBER:

NO CODE LETTER INDICATES DOVETAIL SLOT RECESS PER NASM33750.
“R” INDICATES OFFSET CRUCIFORM RECESS PER NASM14191.

LUBRICATION CODE FOLLOWS SECOND DASH NUMBER:

“L” INDICATES DRY FILM LUBRICANT PER ~~MIL-PRF-46010~~ AS5272, TYPE I.

DRILL CODE FOLLOWS SECOND DASH NUMBER:

“D” INDICATES DRILLED THREADS. /5/

OVERSIZE CODE FOLLOWS SECOND DASH NUMBER:

“X” OR “Y” INDICATE REPLACEMENT BOLT.

WHEN MORE THAN ONE CODE LETTER IS USED FOLLOWING SECOND DASH NUMBER, ARRANGE LETTERS ALPHABETICALLY.

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SHEET 3

EXAMPLES OF PART NUMBER:

NAS1790-4-10	= BOLT; 2500-28 THREAD; .625 GRIP; ALLOY STEEL; CADMIUM PLATED, DOVETAIL SLOT RECESS PER NASM33750.
NAS1790C4-10P	= BOLT; .2500-28 THREAD; .625 GRIP; CRES; CADMIUM PLATED, DOVETAIL SLOT RECESS PER NASM33750.
NAS1790V4-10L	= BOLT; .2500-28 THREAD; .625 GRIP; TITANIUM; DRY FILM LUBRICATED, DOVETAIL SLOT RECESS PER NASM33750.
NAS1790C4-10D	= BOLT; .2500-28 THREAD; .625 GRIP; CRES; DRILLED THREAD, DOVETAIL SLOT RECESS PER NASM33750.
NAS1790-4R10	= BOLT; 2500-28 THREAD; .625 GRIP; ALLOY STEEL, CADMIUM PLATED; OFFSET CRUCIFORM RECESS PER NASM14191.
NAS1790C4-R10B	= BOLT; .2500-28 THREAD; .625 GRIP; CRES; PASSIVATED PER AMS2700, METHOD 2 - NO POST TREATMENT, OFFSET CRUCIFORM RECESS PER NASM14191

NOTES:

- (1) CONCENTRICITY: CONICAL SURFACE OF HEAD TO “ØD” WITHIN .003 FIM. “ØD” TO THREAD PITCH DIAMETER WITHIN .0045 FIM FOR NAS1790-3 THRU NAS1790-5, AND WITHIN .006 FIM FOR NAS1790-6 AND LARGER.
- (2) SURFACE ROUGHNESS: CONICAL SURFACE OF HEAD, HEAD TO SHANK FILLET RADIUS, SHANK AND ALL THREAD ELEMENTS, 32 MICROINCHES Ra; ALL OTHER SURFACES, 125 MICROINCHES Ra PER ASME B46.1.
- (3) DIMENSIONS IN INCHES AND APPLY AFTER FINISH AND PRIOR TO THE APPLICATION OF LUBRICATION UNLESS OTHERWISE SPECIFIED.
- /4/ FOR CRES ONLY, THE EFFECT OF COLD WORK AND AGING INDUCED DURING THE MANUFACTURING CYCLE MAY INCREASE THE ULTIMATE TENSILE STRENGTH OF THE FINISHED PART, BUT ANY PART (TYPE I, II OR III AS LISTED IN NAS4003) SHALL NOT EXCEED 1.3 TIMES THE SPECIFIED TYPE I MINIMUM TENSILE VALUES AS LISTED IN NAS4003 TABLE III.
- /5/ IF REQUIRED, TENSILE TESTING OF BOLTS REQUIRING CROSS-DRILLED THREADS SHALL BE PERFORMED PRIOR TO DRILLING AND THE APPLICATION OF PLATING AND/OR COATINGS. WHEN PARTS HAVE BEEN DRILLED, STRENGTH MAY BE VERIFIED BY SHEAR TESTING, IN LIEU OF TENSILE TESTING, IN ACCORDANCE WITH MIL-STD-1312. USERS SHOULD BE AWARE THAT FASTENERS WITH CROSS-DRILLED THREADS MAY EXHIBIT A REDUCTION IN TENSILE STRENGTH.
- /6/ GRIP LENGTH IS FROM TOP OF BOLT HEAD TO END OF FULL CYLINDRICAL PORTION OF SHANK.
- /7/ “ØA”, “E”, AND “(H)” ARE INCLUDED FOR ENGINEERING REFERENCE PURPOSES ONLY AND ARE NOT TO BE USED FOR INSPECTION. VALUES “ØA”, “E”, AND “(H)” ARE CALCULATED LIMITS RESULTING FROM TOLERANCES ON “ØB”, “ØD”, “M”, “ØV”, AND HEAD ANGLE.
- /8/ DIMENSIONS FOR “M” GAGE PROTRUSION SHALL BE INSPECTED PER NAS9800.
- /9/ HEAD EDGE OUT OF ROUNDNESS SHALL NOT EXCEED “Y” FIM.
- /10/ SHANK SHALL BE STRAIGHT WITHIN “Z” VALUES FIM PER INCH OF LENGTH.
- /11/ POINT SHALL BE FLAT AND CHAMFERED. CHAMFER PLUS INCOMPLETE THREADS SHALL NOT EXCEED “(U)”.

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- /12/ BOLT RECESS SHALL BE TORQUE TESTED IN BOTH INSTALLATION AND REMOVAL DIRECTION WITH APPLICABLE MS14191 OR MS33750 DRIVER. APPLY TABULATED TORQUE IN CW DIRECTION WITH MAXIMUM END LOAD OF 20 POUNDS. PARTS ARE ACCEPTABLE IF RAISED METAL AT EDGE OF RECESS DOES NOT EXCEED TABULATED VALUES. APPLY TABULATED TORQUE IN CCW DIRECTION WITH MAXIMUM END LOAD OF 40 POUNDS. PARTS ARE ACCEPTABLE IF TORQUE IS ATTAINED, WITHOUT RESTRICTION ON RAISED METAL. RECESS CONFIGURATIONS MAY EXHIBIT DIFFERENT PERFORMANCE, BUT BOTH CONFIGURATIONS WILL MEET THE MINIMUM REQUIREMENTS STATED HEREIN. SAMPLING SHALL BE PER ANSI/ASQ Z1.4, 4% AQL LEVEL S-1.
- (13) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
- (14) UNLESS OTHERWISE SPECIFIED, PART INVENTORY MANUFACTURED TO PREVIOUS REVISIONS OF THE APPLICABLE DRAWING OR SPECIFICATION MAY BE PROCURED AND USED UNTIL STOCK IS DEPLETED.
- (15) UNLESS OTHERWISE SPECIFIED HEREIN, REFERENCED DOCUMENTS SHALL BE THE ISSUE IN EFFECT ON DATE OF MANUFACTURE. HOWEVER, EXISTING MATERIAL INVENTORY CERTIFIED TO A PREVIOUS REVISION OF THE APPLICABLE MATERIAL SPECIFICATION(S) IS ACCEPTABLE FOR USE UNTIL DEPLETION.
- (16) DIMENSIONING AND TOLERANCING PER ANSI Y14.5M-1982.

PROCUREMENT SPECIFICATION:

~~MIL-B-87114~~ NAS4002 FOR ALLOY STEEL, NAS4003 FOR CRES, AND NAS4004 FOR TITANIUM.

- ⑥ FATIGUE STRENGTH NOT APPLICABLE TO ALLOY STEEL AND CRES. FATIGUE STRENGTH PER NAS621 FOR TITANIUM.

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC
1000 WILSON BLVD.
ARLINGTON, VA 22209

THIS DRAWING SUPERSEDES ALL ANTECEDENT STANDARD DRAWINGS FOR THE SAME
PRODUCT AND SHALL BECOME EFFECTIVE NO LATER THAN SIX MONTHS FROM THE LAST
REVISION DATE.

FORM 12-01

REVISION
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NAS1790
SHEET 5

HEAD MARKINGS PER SHEET 1 PLUS
OVERSIZE DESIGNATION AS
APPLICABLE (LETTER "X" OR "Y")

INCOMPLETE THREADS
FOR .0158 OVERSIZE
(2PITCHES + .017) MAX
FOR .0312 OVERSIZE
(2PITCHES + .033) MAX

ØD

H

THREADS PER AS8879

Not for Resale, 01/10/2025 22:42:53 MST